

Hadero Besufekad Tadele

Nationality: Ethiopia. Address: 1 Dongxiang Road, Chang'an District, Xi'an Shaanxi, 710129, P.R.China.

☎ (+86) 132-2800-2068 | ✉ haderob4@mail.nwpu.edu.cn | 🌐 hadero-besufekad

Research Interests

My research interests lie in **machine learning, visual perception, and multimodal intelligence** for autonomous systems in complex real-world environments. My work combines **deep learning and geometric vision**, with research spanning visual SLAM, Gaussian Splatting, monocular depth estimation, and large-scale 3D mapping. These experiences led me to focus not only on accurate and efficient scene understanding, but also on systems that support higher-level reasoning and decision-making. I am therefore especially interested in **Vision-Language-Action models** as a natural progression from **perception toward embodied intelligence**. In particular, I aim to study their evaluation in autonomous driving through photorealistic simulation, learned scene representations, and interpretable multimodal benchmarks.

Publications

1. **Besufekad T. Hadero**, Abdulmoiz Ahsan, Dongdong Li, and Tao Yang. (2025) Beyond Implicit Representations: Exploring Gaussian Splatting for Next-Generation SLAM, Introduction and Review *IEEE Internet of Things Journal IF 8.9*, DOI: 10.1109/JIOT.2025.360386, Accepted
2. **Besufekad T. Hadero**, JiaHao Liu, Tao Yang, Member, IEEE, Negaar Rezvanfar, and Ke Ma. FFS-SLAM: Feed-Forward-Enhanced Submap Gaussian SLAM *Under review*,

Experience

COLMAP Optimization for Efficient 3D Reconstruction

Xi'an, China

INTERNSHIP

Oct, 2025 - Present

- Spearheading the optimization of a 3D reconstruction pipeline to significantly reduce the time required to build Gaussian Splatting-based 3D scenes.
- Testing and integrating handheld scanning devices into the pipeline, leveraging expertise in ROS and Gaussian Splatting reconstruction
- Performing parameter tuning and optimization of COLMAP by collecting and analyzing causal video datasets, with a focus on improving performance in challenging scenarios.

Education

Northwestern Polytechnical University

Xi'an, China

M.S. CANDIDATE, ROBOT VISION LAB, COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2023 - Jun. 2026

- Advisor: Prof. Yang Tao
- GPA: 91.8/100 Ranking: **3/29**
- Research Topic: SLAM, Three-dimensional visual perception, Computer Vision

University of Electronic Science and Technology of China

Chengdu, China

B.S., MECHATRONICS (WITH HONORS)

Sep. 2017 - July. 2021

- GPA: 3.78/4.0
- Thesis: Fault Diagnosis Based on Vibration Analysis for Rolling Bearing

Research Experience

Enhancing SLAM with Deep Learning and Neural Scene Representations for Drones

Xi'an, China

THESIS WORK

Sep, 2023 - Present

- Proposed a method to Enhance drone SLAM by introducing novel scene representations and using rendered novel-view images to strengthen relocalization on resource-constrained platforms.
- Integrated deep learning models for core perception tasks, including a convolutional network for dense optical flow prediction between consecutive video frames and a monocular depth estimation network to infer 3D scene geometry from single images.
- Partition the global map into bounded submaps and stream/offload inactive chunks to keep memory flat.
- Use global visual descriptors with neural retrieval, geometric verification, and pose-graph optimization to detect/confirm loops and maintain global consistency.

Project Experience

Monocular 2D-to-3D Content Pipeline for VR

Xi'an, China

TEAM MEMBER

Sep, 2024 - Present

- Built an end-to-end monocular 2D→3D pipeline that turns single images, prerecorded videos, and live camera streams into stereo content for head-mounted displays.
- Applied state-of-the-art depth-prediction networks to produce dense depth maps, enabling accurate scene geometry reconstruction from 2D image.
- Implemented an OpenGL warping pipeline that generates the right-eye view from the source image using the predicted depth and adjustable camera baseline, suitable for real-time playback.
- Used ProPainter to fill the “missing pixels” that appear when generating the right-eye view, and built a small dataset to fine-tune the model—resulting in cleaner fills and fewer visual glitches.

Visual Odometry Pipeline Development Using KITTI Dataset for 3D Vehicle

Xi'an, China

Motion Tracking

TEAM LEADER

Jan, 2024 - Mar, 2024

- Built a visual odometry pipeline on the KITTI driving dataset to estimate 3D vehicle motion from consecutive camera frames.
- Detected visual features, tracked them frame-to-frame, and estimated camera/vehicle pose to recover the trajectory.
- ORB features, KLT optical-flow tracking, and a PnP pose solver implemented with OpenCV.
- Produced stable 3D trajectories and demonstrated an end-to-end perception workflow on real-world data.

Wearable Technologies for Healthcare in Sports

Xi'an, China

TEAM MEMBER

March, 2023

- Built a smartphone-sensor-based activity-recognition demo to show how wearable data and AI can help prevent injuries, enhance performance, and support sports healthcare.
- Collected motion data with the phone's sensors and trained a lightweight deep-learning model to predict common activities.
- Built a Streamlit web app to visualize activity predictions and basic metrics from recorded sessions.

Cityscape Modeling Project: From Drone Capture to 3D Visualization

Xi'an, China

TEAM MEMBER

Dec, 2023

- Built end-to-end 3D city models from UE4/AirSim simulated drone captures: scripted down-looking flights at 100 m AGL, exported frames, reconstructed with MVG/MVS photogrammetry, and published interactive views in Cesium.
- Replicated the workflow on real NWPU campus flights, demonstrating adaptability across environments and datasets.

Awards

HONORS

2026 **Outstanding Graduate**, Outstanding graduates of NWPU 2026

Xi'an, China

2020 **3rd Prize**, UESTC award for Academic Achievement

Chengdu, China

2018 **3rd Prize**, UESTC award for Academic Achievement

Chengdu, China

2016 **Excellent Student**, Adama Science and Technology University Excellent Student award

*Adama,
Ethiopia*

SCHOLARSHIPS

2017 **Betre-Science Scholarship provided by Ethiopia Government (Top 1%)**, Ministry of Science and Technology

*Addis Abeba,
Ethiopia*

2021 **SUSTech School Scholarship 1st prize**, SUSTech University

*Shenzhen,
China*

2023 **CSC Scholarship 1st prize**, The Ministry of Education of China

Xi'an, China

CERTIFICATES

2021	Advanced Learning Algorithms , Machine learning Specialization	<i>Coursera</i>
2021	Supervised Machine Learning: Regression and Classification , Machine learning Specialization	<i>Coursera</i>
2020	Deep Learning Workshop Completion Certificate , Workshop	<i>UESTC</i>

Skills

Programming	Python, C++, C
Libraries	OpenCV, PyTorch, Eigen, CMake, Overleaf - LaTeX
Applications	Linux, Colmap, Meshlab, Unreal Engine(UE4), Pytorch, Adobe Photoshop
Equipments	Drone, VR head, Arduino
Languages	English (Professional proficiency), Amharic, Chinese (Basic proficiency)